Binary sum-product BP

Binary log-likelihood ratio under a depolarizing CSS marginal prior.

v1 : $K = 25$, v2 : $\varepsilon = 10^{-10}$, $d = 0.1$

v1 check: ordered box-plus, clipped after each pair.

$C_K(x) = \max(-K, \min(K, x))$

v1

CHECK UPDATE

$$a \boxplus b = 2\operatorname{atanh}[\tanh(a/2)\tanh(b/2)]$$

$$R_{c \to v} = (-1)^{s_c} \mathcal{B}_K(r_{\partial c \setminus v})$$

POSTERIOR

$$B_v = C_K\Big(\lambda_v + \sum_{c \in \partial v} R_{c \to v}\Big)$$

OUTGOING MESSAGE

$$m^{\text{new}}_{v \to c} = C_K(B_v - R_{c \to v})$$

Clip the prior and each message to magnitude 25. No damping.

v2

CHECK UPDATE

$$q_{cv} = (-1)^{s_c} \prod_{u \in \partial c \setminus v} \tanh(r_{u \to c}/2)$$

$$R_{c \to v} = 2\operatorname{atanh}[C_{1-\varepsilon}(q_{cv})]$$

POSTERIOR

$$B_v = \lambda_v + \sum_{c \in \partial v} R_{c \to v}$$

OUTGOING MESSAGE

$$m^{\text{new}}_{v \to c} = d\, m^{\text{old}}_{v \to c} + (1 - d)(B_v - R_{c \to v})$$

Keep finite LLRs unclipped. Dampen only the outgoing variable message.

Four-state BP4

Pauli states X, Y, Z; A indicates anticommutation. Edges carry commute/anticommute LLRs.

v1 : $K = 25$, v2 : $\varepsilon = 10^{-10}$, $d = 0.1$

v1 check: ordered box-plus, clipped after each pair.

$C_K(x) = \max(-K, \min(K, x))$

v1

CHECK UPDATE

$$a \boxplus b = 2\operatorname{atanh}[\tanh(a/2)\tanh(b/2)]$$

$$R_{c \to v} = (-1)^{s_c} \mathcal{B}_K(r_{\partial c \setminus v})$$

POSTERIOR

$$B_v^a = C_K\Big(\lambda_v^a + \sum_{c \in \partial v} A_{cv}^a R_{c \to v}\Big)$$

OUTGOING MESSAGE

$$m_{v \to c}^{a,\text{new}} = C_K(B_v^a - A_{cv}^a R_{c \to v})$$

Clip the prior, edge projection, and messages. Ties favor X, then Y, then Z; an initial candidate may stop.

v2

CHECK UPDATE

$$q_{cv} = (-1)^{s_c} \prod_{u \in \partial c \setminus v} \tanh(r_{u \to c}/2)$$

$$R_{c \to v} = 2\operatorname{atanh}[C_{1-\varepsilon}(q_{cv})]$$

POSTERIOR

$$B_v^a = \lambda_v^a + \sum_{c \in \partial v} A_{cv}^a R_{c \to v}$$

OUTGOING MESSAGE

$$u_{v \to c}^a = B_v^a - A_{cv}^a R_{c \to v}$$

$$m_{v \to c}^{a,\text{new}} = d\, m_{v \to c}^{a,\text{old}} + (1-d) u_{v \to c}^a$$

No global clip. Strict decision: I, then X, then Z, else Y. Complete one sweep before stopping.